

Generalizable Dynamics Models for Deformable Objects: Tool-Agnostic Model for Tool Geometry Design

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Abstract

Physics-based simulation of deformable objects is often too inaccurate for complex, contact-rich interactions in real-world settings, while learned simulators often suffer from hallucinations when making predictions outside their training distribution. Here we present an approach for learning deformable object dynamics directly from real-world RGB-D observations, rather than synthetic simulation data, accurate enough to drive the computational design of geometry. Our method uses graph neural networks to learn a single dynamics model spanning a variety of tool-object interactions from RGB-D data alone. Surprisingly, we find that training on a small set of tool geometries yields better in- and out-of-distribution generalization than training on a single tool, and can plausibly predict interactions with tools entirely unseen during training. We further exploit the differentiability of the learned model to optimize tool geometry, creating novel tools to solve a variety of downstream tasks.

CCS Concepts

• **Computing methodologies** → Physical simulation; Shape representations; Neural networks;

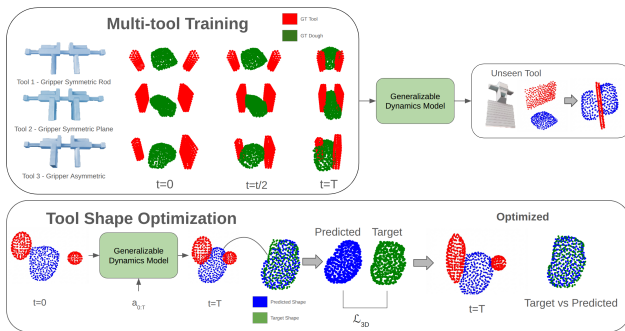


Figure 1: Training and Shape Optimization Pipeline Overview.

1. Introduction

Accurate dynamics models for deformable objects are notoriously difficult to obtain, as analytic and physically based simulators often fail to capture complex, contact-rich interactions in real-world settings. Deep learning offers an alternative [HCM*26], but its value depends on out-of-distribution performance, which prior work leaves unclear. These methods use bespoke models engineered for each tool [SXC*23], train and evaluate on nearly identical distributions [WLGP*23].

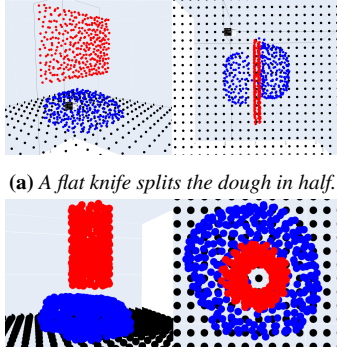
A learned dynamics model also enables tool-geometry optimization through its differentiability. Prior parameterizations, such as

explicit geometry templates [XHS24], rigid links, and cage-based deformation [DHC*25], are limited in expressiveness and restricted to low-dimensional design spaces. We instead represent tool shape with an implicit field, enabling flexible deformation in a high-dimensional space.

In this work, we learn a generalizable dynamics model directly from RGB-D observations that extrapolates far beyond its training distribution, demonstrated on dough manipulation. Given the current dough state and next action, a graph neural network (GNN) predicts the next state. Trained on a variety of tool shapes, our single model generalizes to unseen geometries at test time and, by planning through it, optimizes novel tool shapes to solve a task.

2. Methods

Dynamics Model and Training. We use a GNN to predict future dough states conditioned on robot actions, with tools represented as surface particles. Given the current dough state and an action sequence, the GNN predicts per-particle motions at each timestep and is applied recursively to produce a sequence of dough states. The input point cloud is fused from a multi-view depth camera setup, with no point-to-point correspondences over time. This particle-based representation also lets us train a single dynamics model across different tool shapes, unlike [SXC*23], which trains tool-specific models. Since all tools are 3D-printed from the same material, the representation allows generalization to tool shapes unseen



(a) A flat knife splits the dough in half.

(b) A cylinder tool pushing down at the center of the dough.

Figure 2: Our multi-tool model evaluated on unseen tools.

during training. To improve generalization, we also applied random walk noise to tool particles history before feeding it into our model.

Tool Shape Optimization. We represent tool geometry as points and normals using the Shape-as-Points representation [PJL*21]. The optimized tool points, initial dough points, and action sequence are passed to the GNN to obtain the final dough shape, whose similarity to the target is measured with CD and EMD. The goal is to find a tool shape that, under a fixed action sequence, produces a final dough shape matching the target. Since our GNN is differentiable, we use gradient-based optimization, which scales to large design spaces better than sampling-based approaches. We optimize the tool design with Adam, backpropagating through the reward function after a GNN rollout and through the DPSR framework to update the tool's surface points and normals.

3. Results

In this study, we used real-world data obtained from RoboCook [SXC*23], which consists of rollouts generated from three different tools as shown in Figure 1. We demonstrate our framework's generalization capabilities by examining in-distribution and out-of-distribution cases. In all experiments, we report the average metrics in mm, across 30 rollouts from the test set.

Generalization Experiments. We first study the effect of training a single model on rollouts from multiple tools. Table 1 shows that training on more tools improves shape reconstruction and reduces tool-dough penetration, reflected in the lower SDF value. We then evaluate the model trained on all tools on completely different tools as shown in Figure 2. Although trained only on pushing and pinching tools, it splits the dough with a knife and forms a central hole by pushing a cylinder down.

# Tools	CD ↓	EMD ↓	HD ↓	SDF ↓
1 tool	6.0804	4.6505	16.766	32.898
2 tools	5.7339	4.0243	15.242	29.276
3 tools	5.7354	4.2026	14.359	22.756

Table 1: In-distribution Experiments on GSR.

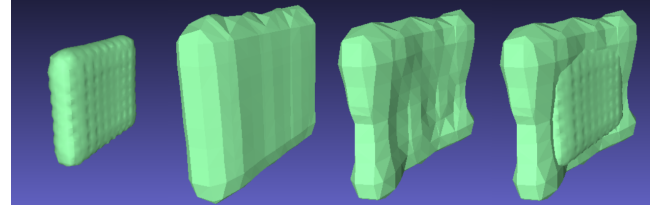


Figure 3: Tool Shape Optimization Results. The columns show ground truth tool, inflated tool, optimized tool shape, and original tool superimposed with optimized tool respectively.

Shape Optimization Experiments. To demonstrate shape optimization, we inflate the ground truth tool and expect the optimized tool to deform back to its original shape. Figure 3 shows the optimized tool shrinking back in the region interacting with the dough, indicating that the GNN gradients are highly informative in the interaction region and usable for local shape optimization.

4. Conclusion

We introduced a graph-based dynamics model for deformable dough manipulation, trained directly from multi-view RGB-D observations. Training a single model on multiple tools, with noise injection into tool particle histories, lets it generalize to unseen tool geometries. On real data, our multi-tool model improves shape reconstruction and reduces tool-dough interpenetration, and qualitative results show out-of-distribution behaviors such as cutting with a knife and punching holes with a cylindrical cutter. Finally, gradients from the model optimize tool shapes in a high-dimensional space through an implicit representation of tool morphology.

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